



Vision Guided Parallel SCARA for Precise pick And place operation

Automated Sorting System

Presenter

Md Roshid Mujaddid [Roll: 2131018]

Group - 3 • 2131014- 2131019





Project Roles

01 **Mechanical**

Md. Roshid Mujadid

Roll: 2131018

02 **Firmware (Kinematics)**

Shavanna Tanjema

Adrita

Roll: 2131016

03 **Firmware (Vision)**

Dibbojoti Dutta

Roll: 2131014

04 **Communication ESP-NOW**

SM Ussas

Roll: 2131019

05 **Calibration & Test**

Siddiqui Hriduon

Roll: 2131015

06 **System Integration**

MD. Tanvir Alam Mollah

Roll: 2131017



Mechanical

01 Linkage Calibration

Measure 5-bar geometric linkages to ± 0.1 mm precision.

02 Encoder Mount

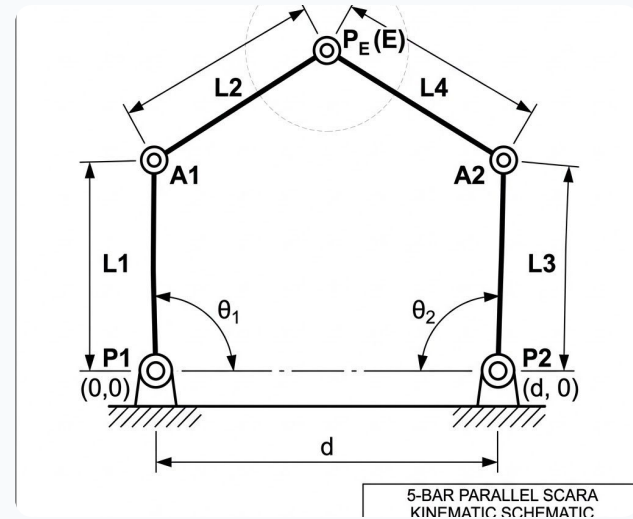
Direct-mount rotary encoders to NEMA 17 shafts (< 0.1 mm runout).

03 EMI Prevention

Isolate encoder signals from power lines to prevent EMI.

04 Signal Integrity

Install decoupling capacitors for clean signal integrity.



Firmware (Kinematics)

01 Inverse Kinematics (IK)

Implement closed-form IK with 2D self-collision monitoring.

02 State Machine

Design an interrupt-driven state machine for non-blocking operations.

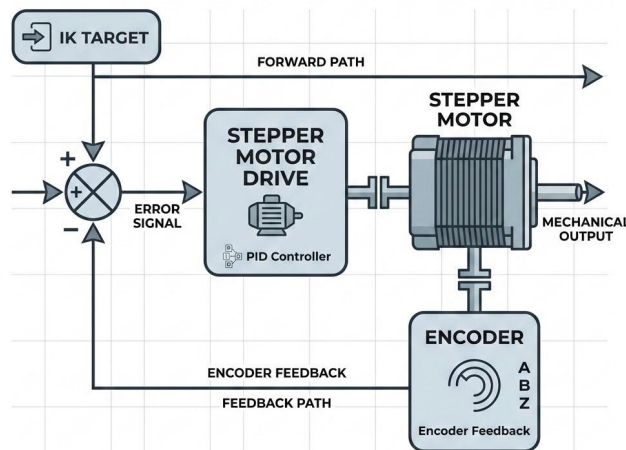
03 Closed-Loop Feedback

Integrate hardware quadrature decoding for closed-loop feedback.

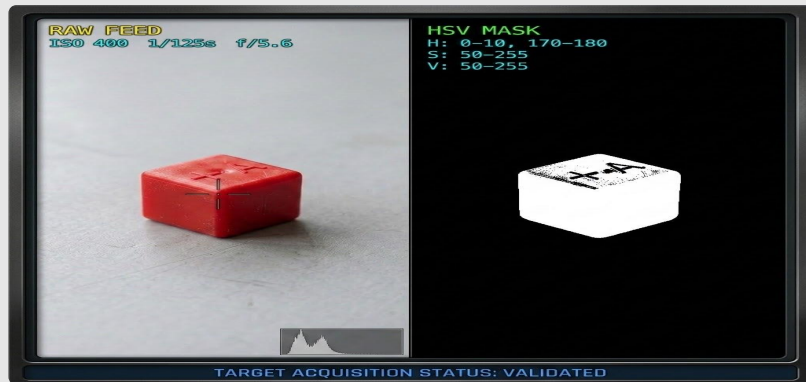
04 Error & Stall Correction

Add position-error correction and automatic motor stall detection.

CLOSED-LOOP STEPPER MOTOR CONTROL SYSTEM

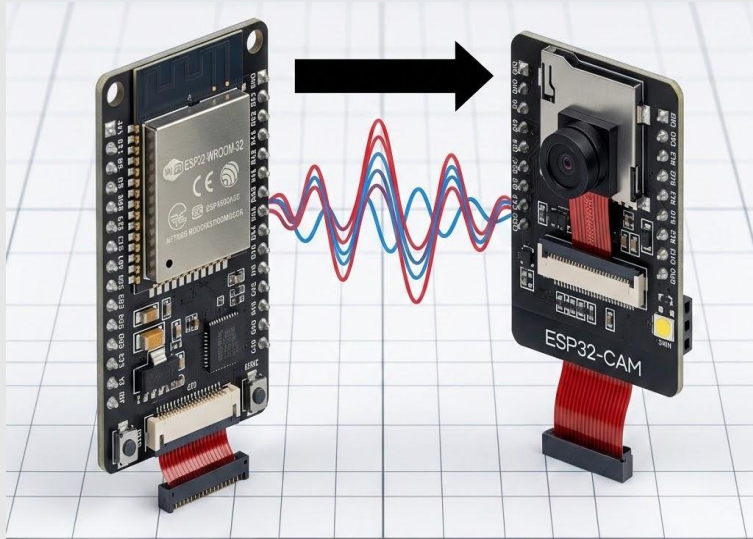


Firmware (Vision)



- **Vision Pipeline:** Deploy a QVGA vision pipeline (RGB565, HSV filter, centroid calculation).
- **Sensor Fusion:** Fuse raw HSV centroids with optical flow frame-differencing.
- **State Estimation:** Apply a 2D Kalman filter for smooth, predictive state estimation.
- **Target Tracking:** Maintain robust target tracking through brief visual occlusions.

Communication (ESP-NOW)



01 Low-Latency MCU

Establish low-latency MCU communication via ESP-NOW.

02 FusionPacket

Transmit 20-byte Fusion Packet (coordinates, velocity, confidence).

03 Data Integrity

Ensure data integrity with CRC-16 checksums and sequence deduplication.

04 Gated Movements

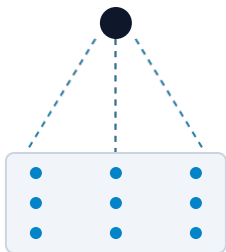
Gate pre-positioning movements using the transmitted confidence score.

Calibration & Validation Infographic

STEP 01

3×3 Grid Calibration

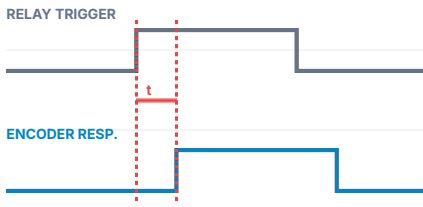
9-point pixel-to-world mapping [u,v]
[x,y] with sub-pixel camera projection.



STEP 02

Timing & Load Test

Dynamic relay response timing under
active magnetic workload.



STEP 03

Workspace Accuracy

Real-time gripper positioning delta
evaluation inside the envelope.



Closed-Loop Encoder Correction

Real-time dynamic feedback loop eliminates accumulative positioning
errors and motor slip under heavy magnetic loads.

OPEN-LOOP ERROR



CLOSED-LOOP ERROR





System Integration & Validation

01 Subsystem Integration

Unite all subsystems into a continuous closed-loop cycle.

02 Power & Watchdogs

Harden power supplies and initialize system watchdog timers.

03 Filter Calibration

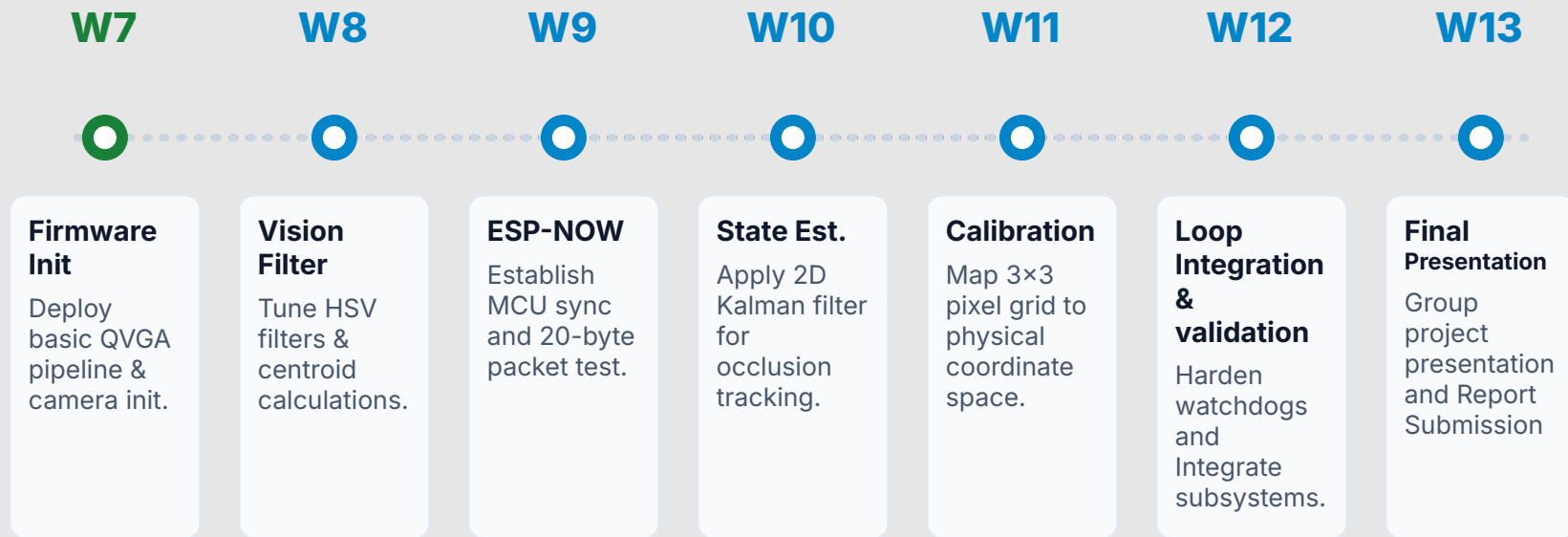
Implement a divergence guard to auto-reset the Kalman filter.

04 Unattended Testing

Execute 30-cycle unattended test (target: ≥ 27 successful cycles).



Project Timeline (Weeks 7-13)





Thank You
